

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0519

Gardner 230 kV
May 21, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0519. The Interconnection Customer proposes a 190.3 MW battery energy storage facility interconnecting at the Gardner 230 kV substation in the Illinois area. The request was received on September 15, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 223 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,475,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gardner-Grand Ridge Battery Projects 230 kV	Top Crop 500-230 kV XFMR	106.1	508	97.1
Gardner-Top Crop 230 kV	Gardner-Grand Ridge Battery Projects Substation 230 kV L	106.0	454	100.0
Palisades 230 kV/1 kV TX	Flood St-Cook Substation 500 kV Line	102.6	806	9.7
Palisades 500/1kV Transformer	Cook-Flood Street 500 kV L	102.2	810	9.7
Grand Ridge Battery Projects-Brevard 230 kV Cir. 1	Grand Ridge Battery Projects-Brevard 230 kV Cir. 2	101.0	801	12.7
Brevard-Grand Ridge Battery Projects 230 kV Line 2	Brevard Substation-Grand Ridge Battery Projects 230 kV	101.0	801	12.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gardner-Grand Ridge Battery Projects 230 kV	Reconductor line	\$3,331,000	37% (\$1,229,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gardner-Top Crop 230 kV	Reconductor line	\$3,552,000	37% (\$1,310,000)
Palisades 230 kV/1 kV TX	Replace transformer with higher-rated unit	\$9,923,000	3% (\$285,000)
Palisades 500/1 kV Auto	Replace transformer with higher-rated unit	\$21,202,000	3% (\$610,000)
Grand Ridge Battery Projects-Brevard 230 kV Cir. 1	Reconductor line	\$276,000	8% (\$21,000)
Grand Ridge Battery Projects-Brevard 230 kV Ckt 2	Reconductor line	\$259,000	8% (\$20,000)
	Total allocated to project		\$3,475,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.